

SATURATED SIDON SETS IN CONSECUTIVE INTERVALS: THE EIGHT-MARK THRESHOLD IS 144

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ABSTRACT. A Sidon set in a finite interval is saturated if no further point of the interval can be added without destroying the Sidon property. Let E_k denote the interval lengths n for which $[n] = \{0, \dots, n-1\}$ contains a saturated k -element Sidon set. We prove that $\max E_8 = 144$. An explicit eight-mark ruler gives the positive case at length 144. To exclude 145, we organize a complete search by minimal witnesses that block the new endpoint and assign every candidate to a canonical search class. The fact that eight marks never reappear is proved separately: any first return after the gap must contain both endpoints. For such rulers, a direct combinatorial count gives at most 155 non-midpoint occurrences; a parity refinement excludes $n \geq 171$, while an exact endpoint search handles $146 \leq n \leq 170$. The accompanying verification archive contains the source code, transcripts, and audit scripts for the finite computations. The proof illustrates how local blocker structure can turn a large exhaustive search into a collection of small, independently checkable cases.

1. INTRODUCTION

Maximal and maximum are different notions. A maximum Sidon set in an interval has as many elements as possible. A maximal Sidon set merely cannot be enlarged while remaining Sidon. We use saturated for this inclusion-maximal property. The distinction leads to a natural spectrum problem: for a fixed interval length n , which cardinalities occur among saturated Sidon subsets?

Unlike maximum cardinality, saturation need not persist when the ambient interval grows. Adding a single new ambient point may make an old saturated set extendible, and there need not be another saturated set of the same size to replace it. For eight marks we determine exactly where this phenomenon becomes permanent: length 144 is possible, length 145 is not, and cardinality eight never returns at a larger length.

We do not claim that $k = 8$ is the first value for which persistence can fail; the smaller cases are not classified here. Our aim is instead to settle the eight-mark case completely and to make the computer-assisted parts of the proof transparent. The key device is the exact blocker criterion of

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Lemma 2.2. It turns saturation, which is a global maximality condition, into a collection of local witnesses. At length 145 these witnesses lead to small endpoint roots and a canonical partition of the search. Beyond 145, a first-return argument and a simple capacity count reduce the infinite problem to one finite endpoint search.

Most of the classical literature on Sidon sets and Golomb rulers studies maximum size, minimum diameter, constructions, or enumeration; see [1, 2, 3, 4, 5]. Here the question is different: the number of marks is fixed, and we ask how long an interval can be while those marks still block every unused point.

The proof follows this division. Section 3 gives a saturated eight-mark ruler of length 144. Sections 4-11 exclude length 145 by a root-based exhaustive certificate. Section 12 then shows that a later return is impossible: any first return must contain both endpoints, a capacity argument rules out all sufficiently large lengths, and an independent exact search closes the remaining finite range.

Theorem 1.1. Let

$$E_k = \{n : [n] = \{0, \dots, n-1\} \text{ contains a saturated } k\text{-element Sidon set}\}.$$

Then $\max E_8 = 144$. Equivalently, $144 \in E_8$ and $n \notin E_8$ for every $n \geq 145$.

The structural statements are kept separate from the numerical audit data. In particular, the manuscript proves why the search classes are exhaustive and disjoint and why the pruning rules are safe; the accompanying programs supply the finite enumerations and the audit scripts check their coverage.

2. SIDON SETS, GOLOMB RULERS, AND BLOCKING

Throughout, $[n]$ means $\{0, \dots, n-1\}$, so n is the number of ambient positions and $n-1$ is the right endpoint. A finite set A of integers is Sidon if all sums $a+b$ with a, b in A and $a \leq b$ are distinct. Equivalently, all positive differences $b-a$ with $a < b$ are distinct. In the latter language A is a Golomb ruler. Write $D(A)$ for its set of positive differences.

Lemma 2.1 (sum-difference equivalence). A finite set A is Sidon if and only if its positive differences are pairwise distinct.

Proof. A nontrivial equality $b-a=d-c$ gives $b+c=d+a$. Conversely, a nontrivial equality of two unordered pair sums can be rearranged to an equality of positive differences. The identical-pair case is the only trivial equality.

A Sidon set $A \subset [n]$ is saturated if $A \cup \{x\}$ is not Sidon for every x in $[n] \setminus A$. We say that such an x is blocked by A .

Lemma 2.2 (exact blocker criterion). Let A be a Golomb ruler, x not in A , and $D=D(A)$. Then x is blocked if and only if either (i) $x = a + d$ or $x = a - d$ for some a in A and d in D , or (ii) $x = (a + b)/2$ for distinct a, b in A with $a+b$ even.

Proof. Before x is inserted, all old positive differences are distinct. Therefore every new violation must involve at least one of the new differences

$|x - a|$. If $|x - a|$ equals an old difference d , then $x = a + d$ or $a - d$. Otherwise a violation can only be an equality between two new differences, $|x - a| = |x - b|$ with $a \neq b$, which is equivalent to $x = (a + b)/2$. Conversely, either condition visibly creates a repeated difference.

2.1. A three-mark example. For $A = \{0, 1, 3\}$, the differences are $\{1, 2, 3\}$. The point 2 is blocked because it is the midpoint of 1 and 3, so the two new distances are both 1. The point 4 is blocked for the other reason: $4 - 1 = 3$ repeats the old distance $3 - 0 = 3$. These are exactly the two mechanisms in Lemma 2.2.

3. LENGTH 144: AN EXPLICIT SATURATED RULER

Consider

$$A_{144} = \{7, 47, 67, 68, 70, 74, 103, 119\}.$$

Its 28 positive differences are

$$D(A_{144}) = \{1, 2, 3, 4, 6, 7, 16, 20, 21, 23, 27, 29, 33, 35, 36, \\ 40, 45, 49, 51, 52, 56, 60, 61, 63, 67, 72, 96, 112\}.$$

They are distinct. A direct application of Lemma 2.2 to the other 136 points of $\{0, \dots, 143\}$ shows that every one is blocked.

Proposition 3.1. A_{144} is a saturated eight-element Sidon subset of $[144]$. Hence $144 \in E_8$.

4. THE 145 PROBLEM AND THE REFERENCE SET F

The interval $\{0, \dots, 144\}$ is only one point larger, but the saturation problem changes. A useful near-saturated Golomb ruler found during the search is

$$T = \{52, 55, 60, 84, 90, 102, 104, 135\}.$$

Exactly one unused point, 12, is addable to T . Reflecting in the midpoint of the ambient interval gives

$$T^* = 144 - T = \{9, 40, 42, 54, 60, 84, 89, 92\}.$$

We use the reflection-invariant reference set

$$F = T \cup T^* = \{9, 40, 42, 52, 54, 55, 60, 84, 89, 90, 92, 102, 104, 135\}.$$

The set F is only an organizing device. It arose from a near-solution and its reflection; no later argument assumes that a genuine solution must resemble T . What matters is the certified reduction below, which shows that any genuine solution could contain at most one point of F .

Proposition 4.1 (global reduction, computer-assisted). Every saturated eight-element Sidon set $A \subset \{0, \dots, 144\}$ satisfies $|A \cap F| \leq 1$.

Section 7 gives the exact partition used to certify this proposition. Each search task there corresponds to a mathematically defined class, so the bookkeeping in the code mirrors a disjoint partition proved in the text.

5. REFLECTION AND SINGLETON CASES

Lemma 5.1 (reflection). The map $r(x) = 144 - x$ preserves the Golomb property and saturation in $\{0, \dots, 144\}$. It also preserves $|A \cap F|$ because $r(F) = F$.

Proof. Reflection preserves every absolute difference. It permutes the ambient interval and maps addable points to addable points. The final assertion follows from the displayed definition of F .

The points of F form seven reflection pairs:

$\{9, 135\}$, $\{40, 104\}$, $\{42, 102\}$, $\{52, 92\}$, $\{54, 90\}$, $\{55, 89\}$, $\{60, 84\}$.

Proposition 5.2 (singleton exclusion, computer-assisted). There is no saturated eight-element Sidon set $A \subset \{0, \dots, 144\}$ with $|A \cap F| = 1$.

All seven reflection pairs are handled by the same partition. For a fixed singleton $s \in F$, there are three endpoint patterns: both endpoints present; exactly one endpoint present; or neither endpoint present.

In the one-endpoint case reflection moves the endpoint to 0, but both singleton orientations are retained. In the no-endpoint case one orientation from each reflection pair is enough. Six orbits were rerun independently in the final audit; the orbit $\{52, 92\}$ is covered by an earlier complete checker using the same Golomb-extension and exact-saturation conditions. Section 8 records the corresponding certificate counts.

Propositions 4.1 and 5.2 imply that every remaining hypothetical solution satisfies $A \cap F = \emptyset$.

6. ENDPOINT-BLOCKING ROOTS

The next definition isolates the smallest reason why the new endpoint 144 can fail to be addable. These minimal endpoint blockers, which we call roots, are the basic units of the exhaustive search.

Definition 6.1 (endpoint-blocking root). Let U be a set of allowed marks contained in $\{0, \dots, 143\}$. An endpoint-144 root is an inclusion-minimal Golomb set $R \subset U$ such that 144 is blocked by R .

Here inclusion-minimal means that 144 is not blocked by any proper subset of R . We order finite subsets increasingly as tuples and use ordinary lexicographic order. Thus, for example, $\{0, 1, 143\}$ is represented by $(0, 1, 143)$.

Lemma 6.2 (root size). Every endpoint-144 root has size two or three.

Proof. Since 144 is larger than every old mark, it cannot be the midpoint of two marks from $\{0, \dots, 143\}$. By Lemma 2.2, blocking 144 therefore means $144 - a = b - c$ for old marks a, b, c with $b > c$. This uses at most three marks. An inclusion-minimal witness cannot have one mark, so its size is two or three.

For instance, $R=\{0, 1, 143\}$ blocks 144 because the proposed new distance $144 - 143 = 1$ repeats the old distance $1 - 0 = 1$.

6.1. Complete root generation. The roots are generated by the following finite procedure. It is separate from the depth-first search used later; its only purpose is to list the minimal endpoint blockers.

ROOTS(U):

```

  L := empty list
  for every 2-subset R of U:
    if R is Golomb and blocks 144,
      and no proper subset blocks 144:
        append R to L
  for every 3-subset R of U:
    if R is Golomb and blocks 144,
      and no proper subset blocks 144:
        append R to L
  sort L lexicographically and return L

```

Lemma 6.3 (correctness of ROOTS). ROOTS(U) returns exactly the endpoint-144 roots contained in U , each exactly once.

Proof. Soundness is immediate from the three tests in the algorithm. For completeness, let R be an endpoint-144 root. Lemma 6.2 gives $|R|$ in $\{2, 3\}$, so R is visited by the corresponding subset loop and passes all tests by definition. A finite subset is visited once, and the final sort only orders the output.

Two instances are used in the proof. For the global reduction, $U=\{0, \dots, 143\}$; the deterministic exhaustive generator returns 5136 relevant minimal endpoint roots, of which 72 are pairs and 5064 are triples. For the final empty- F case, $U = \{0, \dots, 143\} \setminus F$; the corresponding generator returns 3779 roots, 57 pairs and 3722 triples. These four integers are computational outputs of ROOTS(U), not closed-form theoretical counts. Their index ranges and pair/triple decompositions are checked independently by the accounting audits described in Section 11.

6.2. Canonical root assignment. A complete candidate may contain more than one root. We therefore define the assignment mathematically before describing the search.

Definition 6.4 (canonical root). For a Golomb set A with $144 \notin A$ and with 144 blocked, define $\rho(A)$ to be the lexicographically least endpoint-144 root R contained in A .

Lemma 6.5 (existence and uniqueness of rho). Every such A has exactly one canonical root $\rho(A)$.

Proof. By Lemma 6.2, A contains at least one inclusion-minimal endpoint-blocking witness. The family of roots contained in A is finite and nonempty. Lexicographic order is a total order, so it has a unique least element.

The DFS associated with root R may generate a complete ruler A only if $R \subset A$, but it accepts A into R 's certificate class only when $\rho(A) = R$. Consequently the classes $\{A : \rho(A) = R\}$ are pairwise disjoint and their union is the complete search space under the stated endpoint and forced-mark restrictions. This is the formal meaning of “first root” used in the code.

7. CERTIFYING THE GLOBAL REDUCTION

We now prove Proposition 4.1 by partitioning every hypothetical saturated set with $|A \cap F| \geq 2$ into one uniquely determined search class.

7.1. Both endpoints present. If 0 and 144 are both marks, define $\phi(A)$ to be the two smallest elements of $A \cap F$. There are $\binom{14}{2}=91$ possible values. For each unordered pair $P \subset F$, the search forces $\{0, 144\} \cup P$ and enumerates all Golomb-compatible extensions to eight marks. A leaf belongs to the certificate class P only if $\phi(A) = P$. Thus the 91 classes are disjoint and exhaustive.

7.2. At most one endpoint present. If exactly one endpoint is present, reflect when necessary; if neither is present, do nothing. In either case we may assume $144 \notin A$. By saturation, 144 is blocked and $\rho(A)$ is defined. Again let $\phi(A)$ be the two smallest elements of $A \cap F$.

Lemma 7.1 (canonical global key). Every saturated candidate with $144 \notin A$ and $|A \cap F| \geq 2$ has a unique key $(\rho(A), \phi(A))$.

Proof. Both coordinates exist and are unique by Definition 6.4 and by the ordinary order on F . Hence the ordered pair is unique. Conversely every candidate satisfying the hypotheses is assigned to its key, so no candidate can escape the partition.

The root generator returns 72 pair roots and 5064 triple roots in this global search. For the pair-root layer, the implementation materializes all $72 \cdot 91 = 6552$ possible root/ F -pair keys and then applies the canonical-key test. For triple roots, the same phi-test is applied inside the root search. Thus 5136 and 6552 are deterministic enumerative outputs and task counts. The accounting audit verifies root indices 0,...,5135, pair-task indices 0,...,6551, the expected root/ F -pair attached to each task, and saturated count zero throughout.

Proposition 7.2 (certificate conclusion). All classes in Sections 7.1-7.2 have saturated-leaf count zero. Therefore $|A \cap F| \geq 2$ is impossible, proving Proposition 4.1.

8. SINGLETON CERTIFICATES

The singleton case uses the same idea, so we state the partition explicitly rather than treating the archived computation and the later reruns as different mathematical arguments.

Definition 8.1 (singleton search class). Fix a reflection orbit $\{s, 144 - s\} \subset F$. A singleton candidate is a saturated eight-mark Golomb ruler A

with $A \cap F = \{s\}$ or $\{144 - s\}$. Its endpoint type is B if both endpoints occur, O if exactly one occurs, and N if neither occurs.

Reflection identifies the two B orientations and the two N orientations; in type O it normalizes the unique endpoint to 0 but both singleton orientations are retained. Within each endpoint type, endpoint-root or minimum-additional-mark keys are used exactly as in the final search, and an accounting script checks the expected index interval.

Six reflection orbits, $\{9,135\}$, $\{40,104\}$, $\{42,102\}$, $\{54,90\}$, $\{55,89\}$, and $\{60,84\}$, were rerun independently in the final audit. Together these runs contain 85,818,300,741 complete leaves, none saturated.

The remaining orbit $\{52,92\}$ is covered by an earlier complete root certificate using the same Golomb-extension and exact-saturation tests; its 3665-root count is likewise output of a deterministic generator. The verification archive contains both the archived certificate and the later rerun material. Thus the difference is historical, not mathematical.

Proposition 8.2. Every singleton search class has saturated-leaf count zero. Hence $|A \cap F|=1$ is impossible.

9. THE FINAL EMPTY-F CASE

It remains to exclude $A \cap F = \emptyset$.

9.1. Both endpoints present. If $\{0, 144\} \subset A$, the remaining six marks are partitioned by the smallest additional mark. The 128 admissible minimum-mark branches are disjoint by definition and exhaustive. They contain 1,472,636,596 complete leaves and no saturated leaf.

9.2. Endpoint 144 absent. Otherwise, after reflection we may assume $144 \notin A$. The property $A \cap F = \emptyset$ is reflection-invariant because F is reflection-invariant. Set $U = \{0, \dots, 143\} \setminus F$. The deterministic $\text{ROOTS}(U)$ computation returns 3779 endpoint roots; the accounting audit independently verifies that these are 57 pair roots and 3722 triple roots and that the complete index range is present. Definition 6.4 assigns every candidate to exactly one of these root classes. Every class has saturated-leaf count zero.

Proposition 9.1. No saturated eight-element Sidon set in $\{0, \dots, 144\}$ has empty intersection with F.

9.3. Exclusion of length 145. Suppose A were a saturated eight-element Sidon set in $\{0, \dots, 144\}$. Proposition 4.1 gives $|A \cap F| \leq 1$. Proposition 8.2 excludes equality, so $A \cap F = \emptyset$. Proposition 9.1 then gives a contradiction. Thus $145 \notin E_8$. Together with Proposition 3.1, this establishes the adjacent reversal $144 \in E_8$, $145 \notin E_8$. Section 12 rules out every later return and completes Theorem 1.1.

10. SEARCH ALGORITHM AND EXACT SATURATION TEST

The preceding sections explain which candidates must be searched. We now describe the exact test carried out inside one class.

```

EXTEND(partial ruler A, used differences D):
  choose the next mark x in increasing order
  reject x if two new differences |x-a| coincide
  reject x if a new difference lies in D
  otherwise insert x and its new differences
  recurse until eight marks have been chosen
  at eight marks, test every unused point by Lemma 2.2
  record SAT only if no unused point is addable

```

Thus every leaf counted by the certificate is a complete eight-mark Golomb ruler satisfying the forced conditions of its class. Saturation is not inferred from a heuristic score: every unused point is tested by the exact blocker criterion.

10.1. Safe last-mark pruning. A speedup used in the re-audit does not alter completeness. Suppose one mark remains and t is currently addable. For a proposed last mark x , if $x \neq t$ and t remains addable after x is inserted, the resulting eight-mark ruler cannot be saturated and the branch may be discarded. If $x = t$, or t becomes blocked, the exact leaf test is still performed. Hence the pruning removes only provably nonsaturated leaves.

11. AUDIT AND CERTIFICATE ACCOUNTING

The following counts are included so that an independent rerun can be compared with the supplied certificates. They are audit data, not substitutes for the completeness lemmas above: the proof does not depend on a closed formula for the number of roots or tasks.

Claim	Partition	Expected classes	Complete leaves	SAT leaves
144 positive	direct	1 witness	136 unused points checked	0 addable
$ A \cap F \geq 2$, both endpoints	two smallest F -marks	91	318,781,896	0
$ A \cap F \geq 2$, triple roots	$\rho + \phi$	5064	27,491,338,178	0
$ A \cap F \geq 2$, pair roots	$\rho + \phi$	6552 tasks	3,935,570,975	0
Six singleton reruns	orbit/endpoints/root keys	complete	85,818,300,741	0
Singleton $\{52, 92\}$	archived root partition	3665 roots	complete archived run	0
Empty F , both endpoints	minimum added mark	128 branches	1,472,636,596	0
Empty F , 144 absent	ρ	3779 roots	complete root audit	0

For the independent global re-audit, the combined total is 37,173,916,771 DFS nodes and 31,745,691,049 complete leaves. The accounting script verifies exactly 5136 endpoint-root indices 0,...,5135, with 72 pair roots and 5064 triple roots; exactly 6552 pair-root/ F -pair tasks 0,...,6551; all 91 both-endpoint

F-pairs; semantic agreement of repeated checkpoint records; and saturated count zero in every record. The numerical values 5136 and 6552 are outputs of the enumerators and audits; the logical proof uses the exhaustivity of the definitions and the verified coverage, not an independent theoretical formula for these numbers.

For the six independently rerun singleton orbits, the accounting audit verifies every expected seed/root index, no out-of-range or malformed nonempty logs, and no positive saturated count. The seventh orbit $\{52,92\}$ is represented by its archived complete certificate and reproducible source material. In the ancillary verification archive these materials are preserved unchanged under legacy_145, together with their original compact certificate summaries, audit scripts, raw re-audit logs, and SHA-256 manifests.

12. WHY EIGHT MARKS NEVER RETURN AFTER 144

The calculation at length 145 creates the first gap after the positive example at 144. To prove that 144 is truly the last element of E_8 , we must still rule out the possibility that saturated eight-mark rulers reappear later. The next lemma is what makes that question manageable.

12.1. First-return endpoint lemma. Lemma 12.1 (first-return endpoint lemma). Suppose $n \in E_k$ and $n - 1 \notin E_k$. Then every saturated k -element witness in $[n] = \{0, \dots, n - 1\}$ contains both endpoints 0 and $n - 1$.

Proof. Let A be saturated in $[n]$. If $n - 1$ were not a mark, then every point of $[n - 1] \setminus A$ would already be blocked by A , so A would be saturated in $[n - 1]$, contradicting $n - 1 \notin E_k$. If 0 were not a mark, reflect A by $x \mapsto n - 1 - x$. Reflection preserves all absolute differences and saturation, and the reflected witness would omit $n - 1$, reducing to the first contradiction.

Consequently, if E_8 had any element larger than 145, the least such element would be witnessed by an eight-mark Golomb ruler containing both endpoints.

12.2. A blocker-capacity bound. Let

$$A = \{0 = x_0 < x_1 < \dots < x_6 < d = x_7\}$$

be an endpoint-containing eight-mark Golomb ruler, with ambient size $n = d + 1$. We count how many ambient positions such a ruler could possibly cover. The eight marks themselves are already occupied. Every other blocked position is covered by one of the two mechanisms in Lemma 2.2: it is either a midpoint, or one of its new distances repeats an old difference. If

$$|x - a| = |b - c|,$$

then solving for x gives $x = a + (b - c)$ or $x = a - (b - c)$. After relabelling, these are precisely expressions of the forms

$$2a - b, \quad a + b - c, \quad \frac{a + b}{2}.$$

Thus every blocked point is represented by at least one expression counted below. Counting candidate positions with multiplicity can only overestimate the true blocking capacity.

We begin with the two-mark extrapolations $2a - b$. Fix the subtracted mark b . If b is one of the six interior marks, the choices $a = 0$ and $a = d$ give values outside $[0, d]$, while $a = b$ gives only the old mark b . At most the other five interior choices of a can therefore produce new in-range positions. Over the six possible interior values of b , this contributes at most $6 \cdot 5 = 30$ occurrences.

If b is an endpoint, then for each interior center a the two endpoint-based candidates are $2a$ and $2a - d$. At most one can be a new point of $[0, d]$: for $a < d/2$ only $2a$ can lie in the interval, for $a > d/2$ only $2a - d$ can, and for $a = d/2$ the two values are the old endpoints d and 0 . Hence endpoint subtraction contributes at most six more occurrences, and the total pair-extrapolation capacity is at most

$$30 + 6 = 36.$$

Now take three distinct marks $p < q < r$. The three affine candidates of type $a + b - c$ are $p + q - r$, $p + r - q$, and $q + r - p$. The central value $p + r - q$ always lies between p and r , so the $\binom{8}{3} = 56$ triples contribute at most 56 central occurrences.

If p, q, r are all interior, either side value may also lie in $[0, d]$, giving at most

$$2 \binom{6}{3} = 40$$

additional occurrences. For triples involving an endpoint, fix an interior pair $a < b$. The triple $\{0, a, b\}$ can contribute only the right-side value $a + b$, whereas $\{a, b, d\}$ can contribute only the left-side value $a + b - d$. At most one of these lies in $[0, d]$; if $a + b = d$, the two values are merely the old endpoints. Hence endpoint triples contribute at most

$$\binom{6}{2} = 15$$

further side occurrences.

Adding the eight marks and the three non-midpoint contributions gives the universal bound

$$8 + 36 + 56 + 40 + 15 = 155.$$

There are at most $\binom{8}{2} = 28$ midpoint occurrences, since an integer midpoint can only come from a pair of marks of the same parity. The crude total capacity is therefore at most $155 + 28 = 183$ ambient positions. This already rules out endpoint-containing saturated eight-mark rulers for every $n \geq 184$.

Only the finite range $171 \leq n \leq 183$ remains beyond the later exact search. Here a small parity refinement is enough. If r_i is the number of marks congruent to $i \pmod{4}$, then the parity of an integer midpoint is determined

by the two residue classes, and the even and odd midpoint capacities are bounded by

$$M_{\text{even}} = \binom{r_0}{2} + \binom{r_2}{2} + r_1 r_3, \quad M_{\text{odd}} = r_0 r_2 + \binom{r_1}{2} + \binom{r_3}{2}.$$

The non-midpoint expressions can be split by parity as well. The parity of $2a - b$ is the parity of b , and the parity of $a + b - c$ is the parity of the sum of the three marks. Thus, if e and o are the total numbers of even and odd marks, the central triple capacities are

$$\binom{e}{3} + e \binom{o}{2} \quad \text{on even positions,} \quad \binom{o}{3} + o \binom{e}{2} \quad \text{on odd positions.}$$

The all-interior side terms have the analogous bounds, doubled because there are two side expressions per interior triple.

The remaining endpoint terms are handled conservatively. When d is even, both endpoints are even, so their parity contribution is fixed. When d is odd, the endpoints have opposite parity; rather than guessing the split, the audit allows every possible distribution of the six endpoint-subtracted extrapolations and the 15 endpoint-side occurrences between even and odd positions. The script `e8_first_return_residue_bound.py` enumerates every mod-4 count vector of the six interior marks and all of these undetermined splits. Therefore every actual endpoint-containing ruler is dominated by at least one tested profile. Collisions or expressions falling outside $[0, d]$ can only lower the true capacity.

The resulting feasible-profile counts are:

ambient n	168	169	170	171–183
feasible residue profiles	46	2	12	0 for every n

Proposition 12.2 (capacity exclusion). No endpoint-containing saturated eight-element Sidon set exists in $[n]$ for any $n \geq 171$.

Proof. The preceding count covers every blocked point by Lemma 2.2 and gives an upper-bound profile for every possible endpoint-containing ruler. The residue audit finds no feasible profile for $171 \leq n \leq 183$. For $n \geq 184$, the crude bound 183 is already smaller than the ambient size n . Hence no endpoint-containing saturated eight-mark ruler exists for any $n \geq 171$.

12.3. Exhaustion of the finite first-return range. The capacity argument leaves only $146 \leq n \leq 170$. In this range a direct endpoint-containing search is feasible. The program `e8_firstreturn_146_170.cpp` fixes 0 and $d = n - 1$, enumerates the six interior marks in increasing order, and normalizes reflection by requiring $x_1 + x_6 \leq d$. Every endpoint-containing ruler is therefore represented once, apart from reflection-fixed ties, which are retained once.

The search uses the same capacity bound only as a safe pruning device. At a partial state it subtracts blocker expressions already forced outside the

interval, gives the unchosen marks the most optimistic possible midpoint contribution, and lower-bounds further unavoidable losses. A branch is discarded only when even this optimistic completion cannot cover all n ambient positions. Every complete ruler that survives the pruning is then tested for saturation point by point using Lemma 2.2.

The fresh unified run for all 25 lengths gives:

n range	lengths	DFS nodes	capacity-surviving rulers	SAT
146–150	5	2,487,950,719	731,667,499	0
151–155	5	1,987,073,077	515,048,548	0
156–160	5	1,324,929,800	326,272,590	0
161–165	5	802,137,417	186,979,832	0
166–170	5	340,771,773	62,869,651	0
total	25	6,942,862,786	1,822,838,120	0

The complete transcript `e8_firstreturn_146_170.log` contains one row for every $n = 146, \dots, 170$. The independent audit script verifies the exact index range, parses every row, and checks saturated count zero. In total the run visits 6,942,862,786 DFS nodes and exact-tests 1,822,838,120 capacity-surviving complete rulers.

Proposition 12.3 (finite first-return exclusion). No endpoint-containing saturated eight-element Sidon set exists in $[n]$ for $146 \leq n \leq 170$.

12.4. Completion of the threshold theorem. Proof of Theorem 1.1. Proposition 3.1 gives $144 \in E_8$ and Section 9 gives $145 \notin E_8$. Suppose some $n \geq 146$ belonged to E_8 , and choose the least such n . Then $n - 1 \notin E_8$, so Lemma 12.1 forces both endpoints into every witness at n . If $n \leq 170$ this contradicts Proposition 12.3; if $n \geq 171$ it contradicts Proposition 12.2. Hence no such n exists and $\max E_8 = 144$.

13. DATA AND CODE AVAILABILITY

The ancillary archive `Huber_Saturated_Sidon_E8_Threshold_144_Verification_v5.zip` contains the material needed to reproduce the computer-assisted parts of the proof. The directory `legacy_145` preserves the complete length-145 certificate package. The directory `first_return` contains the endpoint search for $n = 146, \dots, 170$, its full transcript, an independent transcript audit, and the parity/mod-4 capacity check. The `oeis` directory contains the companion b-file through 183, a small-spectrum Maple enumerator, and explicit 15-mark witnesses for $n = 178, \dots, 183$ with an exact verifier.

The large root, task, and leaf counts reported above are outputs of deterministic programs and are checked by the supplied audits. The logical role of the manuscript is different: it proves which finite families have to be generated, why the canonical keys cover them without loss, and why each

pruning rule preserves completeness. In particular, the constants 36, 40, 15, and 155 in Section 12.2 are derived in the text rather than assumed from a program.

14. DISCUSSION

Theorem 1.1 is stronger than an adjacent reversal: it determines the exact terminal ambient length for saturated eight-element Sidon sets. The length 144 is attainable, length 145 is not, and the first-return argument proves that cardinality 8 never reappears. This sharply distinguishes saturation from the classical extremal problem, where an old Sidon set remains available when the ambient interval grows.

There are two ideas in the proof that seem useful beyond this particular value of k . Near a difficult endpoint, the blocker criterion reduces saturation to small local witnesses, which can be organized into canonical roots. After a gap has been found, the first-return lemma forces both endpoints and makes a coarse capacity count surprisingly effective. In the present problem those two ideas reduce an apparently open-ended search to a finite collection of auditable computations.

The numerical counts are included for reproducibility, not as a substitute for the argument. What carries the proof is the exhaustivity of the search partitions, the exact saturation test at every surviving leaf, the first-return endpoint lemma, and the capacity bounds used in the no-return step.

15. OPEN QUESTIONS

1. How complicated can E_k be for fixed k other than 8? In particular, when can it have internal gaps, and can the same cardinality return more than once after disappearing?
2. Can exact terminal values $\max E_k$ be determined for infinitely many k , or bounded structurally without large exhaustive searches?
3. Can the blocker-capacity method be generalized to obtain effective upper bounds for a first return in E_k for arbitrary fixed k ?
4. Can the combination of near-saturated rulers, endpoint roots, and residue-capacity profiles lead to a general theory of saturated Sidon spectra?

ACKNOWLEDGEMENTS AND COMPUTATIONAL DISCLOSURE

The project used computer-assisted exploration, followed by independent reruns and accounting audits for the final certificates. The ancillary verification archive contains the source code, raw logs, audit scripts, manifests, and OEIS companion data. The completeness arguments themselves are stated in the manuscript; the ancillary archive supplies the finite computations to which those arguments are applied.

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APPENDIX A. FORMAL CERTIFICATE INVARIANTS

For reference, we collect the invariants checked by the verifier. They also give a concise specification for an independent implementation.

- Golomb invariant: every partial state has pairwise distinct positive differences.
- Ordering invariant: marks are inserted in increasing order, so each completed set is generated once inside a fixed forced class.
- Root invariant: a root is accepted by $\text{ROOTS}(U)$ exactly when it is Golomb, blocks 144, and no proper subset blocks 144.
- Canonical-root invariant: a completed candidate belongs only to the class indexed by $\rho(A)$.
- Global-pair invariant: when $|A \cap F| \geq 2$, $\phi(A)$ is the two smallest F-marks and is unique.
- Leaf invariant: $\text{SAT}=1$ is recorded only after every unused ambient point fails the exact addability test.
- Accounting invariant: every expected class index occurs, every recorded key equals the key prescribed by its index, and all duplicate checkpoint records agree.

First-return invariant: if $n \in E_k$ but $n - 1 \notin E_k$, every witness at n contains both endpoints.

Reflection invariant for the return search: every endpoint-containing ruler is represented with $x_1 + x_6 \leq d$.

Capacity invariant: a pruned partial state is discarded only when an optimistic upper bound on still-coverable ambient positions is smaller than n .

Return-leaf invariant: every complete capacity-surviving ruler in the range $146 \leq n \leq 170$ is tested by the exact saturation predicate, and every transcript row has saturated count zero.

These invariants separate the mathematical completeness question from the numerical size of the computation. The legacy invariants plus the new first-return invariants imply Theorem 1.1 once the supplied audits return the stated zero saturated counts and zero feasible residue profiles.

APPENDIX B. OEIS COMPANION SPECTRUM DATA THROUGH 183

This appendix records the consequence used for the companion OEIS submissions A399118 and A399119. It is not needed for Theorem 1.1. Define $T(n, k)$ as the number of inclusion-maximal Sidon subsets of $\{1, \dots, n\}$ of cardinality k , let $S(n) = \{k : T(n, k) > 0\}$, and set $a(n) = |S(n)|$.

Previously certified spectrum computations provide the lower support edge through 144 and constructive coverage for the larger cardinalities. The ranges needed here include E_9 containing $[45, 183]$, E_{10} containing $[56, 183]$, E_{11} containing $[73, 183]$, E_{12} containing $[86, 183]$, E_{13} containing $[107, 183]$, and E_{14} containing $[128, 183]$. Theorem 1.1 supplies the new fact that 8 is absent for every $n \geq 145$.

For $152 \leq n \leq 177$ the maximum Sidon size is 15, so a maximum set is automatically inclusion-maximal. For $178 \leq n \leq 183$ the maximum size is 16; explicit saturated 15-mark witnesses are included in the ancillary archive and independently verified, while maximum-size 16-sets are again automatically inclusion-maximal. Thus $S(n)$ is an interval of consecutive integers for every $1 \leq n \leq 183$.

n range	$\min S(n)$
1	1
2–4	2
5–10	3
11–22	4
23–42	5
43–67	6
68–101	7
102–144	8
145–183	9

The upper support edge is A143824(n). Consequently, for $1 \leq n \leq 183$,

$$a(n) = A143824(n) - A382397(n) + 1.$$

The ancillary file `oeis/b399119.txt` records the resulting 183 terms; `oeis/Saturated_Sidon_Spectrum_Maple.mpl` provides an independent exact enumerator for the initial rows.

INDEPENDENT RESEARCHER